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# Lifted Cavity Networks: In-Context Sensor Fusion with Missing Sensors and Shared Noise

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## Abstract

1 Low-cost sensor deployments must estimate a target from whichever sensors work,  
2 after a short calibration, while temperature, humidity and drift perturb several  
3 sensors at once. We cast this as in-context probabilistic prediction that must remain  
4 correct for every missingness pattern and any number of sensors. Products of  
5 per-sensor experts over the target meet both requirements, but we prove they cannot  
6 be Bayes-consistent when sensors share noise. We propose *lifted cavity networks*:  
7 each observed sensor contributes a rank-one Gaussian site over the target and a  
8 low-dimensional shared nuisance, exact for factor-structured noise under every  
9 pattern once the nuisance dimension reaches the noise rank, which we show is also  
10 necessary. The sites start from a factor analysis of the incomplete calibration rows  
11 and are refined by a learned, cavity-conditioned re-linearization. On synthetic tasks  
12 with exact and Bayes-optimal references, an 8k-parameter lifted cavity network  
13 closes 58% of the gap between the closed form and the Bayes-optimal predictive  
14 with two sensors missing, a pattern unseen in training, against 34% for a 25-times-  
15 larger TabPFN-v2-style transformer trained on the same tasks, and it transfers from  
16 five sensors to sixteen. After fine-tuning on real sensor networks it improves on  
17 the closed forms, but the transformer is better still. As the transformer’s output  
18 layer, lifted sites beat the plain transformer on fresh synthetic tasks (by 0.035 nats  
19 with five sensors, 0.150 with sixteen) and recover 75% of the real-data gap on a  
20 new target, although non-inferiority to the fine-tuned transformer is not established.  
21 All three evaluation protocols were committed before their test data existed, and  
22 every endpoint is reported.

## 23 1 Introduction

24 Low-cost sensors now carry much of the world’s environmental, industrial and clinical measurement.  
25 A typical deployment co-locates a few cheap sensors with a reference instrument for a short calibration  
26 period, and from then on predicts the reference quantity from the sensors alone [38]. Three features  
27 make this hard. First, every deployment is different: gains, cross-sensitivities and noise depend  
28 on the unit, its placement and its age, and the calibration record is short. Second, sensor noise is  
29 *shared*: temperature, humidity, local emissions and drift move several sensors at once, so sensors are  
30 not conditionally independent given the target. Third, sensors fail, are serviced or are added, so at  
31 prediction time an arbitrary subset is available, often one never seen together during calibration.

32 We study this as *in-context* probabilistic prediction. A network receives a deployment’s labeled  
33 calibration rows (the support), with whatever entries are missing, and a query made of an arbitrary  
34 subset of sensor readings, and it returns a predictive distribution for the target. Trained once across  
35 many deployments, such a network adapts to a new one from its calibration rows alone, as conditional  
36 neural processes and prior-fitted networks do for regression and tabular prediction [8, 26, 15].

The problem imposes two structural requirements. The predictor must be *mask-consistent*: defined and accurate for every subset of observed sensors, including subsets absent from training. It must also be *width-agnostic*: usable for any number of sensors. A product of per-sensor experts over the target meets both by construction, because a missing sensor simply contributes no factor [13, 35]. Learned message passing in the style of expectation propagation (EP) refines such experts using context from the other sensors [24, 12].

### Contributions.

1. **When products of experts fail (§3–4).** Under shared noise, no product of mask-independent experts over the target alone is Bayes-consistent across missingness patterns; the optimal weight of a sensor can even change sign with the other sensors present. More generally, static Gaussian sites over the target and  $K'$  nuisance dimensions can be consistent for every pattern only if  $K'$  reaches the rank of the shared noise, and then they are exact. An EP-style network with scalar cavities fails accordingly in practice.
2. **Lifted cavity networks (§4).** Each observed sensor contributes a rank-one Gaussian site over the target and a low-dimensional shared nuisance, and missing sensors are omitted exactly. The sites start from a factor analysis of the incomplete calibration rows and are refined by a small network that re-linearizes each sensor’s response under its cavity.
3. **A pre-registered evaluation (§6)** against exact posteriors, a Hamiltonian Monte Carlo estimate of the Bayes-optimal predictive, tuned closed forms and a TabPFN-v2-style transformer trained on the same task distribution, on synthetic panels and two real sensor networks. Three protocols were committed before their test data existed, and every endpoint is reported.

**What we find.** Structure buys sample efficiency and extrapolation, but on real data it costs flexibility.

- On synthetic tasks, the 8k-parameter lifted cavity network beats a 203k-parameter TabPFN-v2-style transformer trained on the same tasks for every number of missing sensors, closing 58% of the achievable gap against 34%; its margin grows on 16 sensors after training on five. It loses its edge under strong response nonlinearity and on linear sensors (§6.3).
- On two real sensor networks, the fine-tuned lifted network improves on the closed forms, but a fine-tuned transformer is better, and one pre-registered endpoint fails (§6.5).
- As the transformer’s output layer (§6.6), lifted sites beat the plain transformer on fresh synthetic panels and the lifted network on a new real target, recovering 75% of the real-data gap; non-inferiority to the fine-tuned transformer is not established.

## 2 Problem setup

A task (deployment)  $\tau \sim p(\tau)$  defines a joint distribution  $p_\tau(x, y)$  over  $P$  sensor readings  $x \in \mathbb{R}^P$  and a scalar target  $y$ . The support  $\mathcal{D} = \{(x_i, m_i, y_i)\}_{i=1}^n$  holds  $n$  labeled rows whose entry  $x_{ij}$  is observed where  $m_{ij} = 1$ . A query is a reading  $x_S$  of an observed subset  $S \subseteq \{1, \dots, P\}$ . The target of prediction is the posterior predictive

$$p(y | x_S, \mathcal{D}) = \int p_\tau(y | x_S) p(\tau | \mathcal{D}, x_S) d\tau, \quad (1)$$

and methods are scored by the negative log-likelihood (NLL) of the held-out target. Two privileged references bound what is possible on synthetic tasks: the *oracle*  $p_\tau(y | x_S)$ , which knows the task, and the *Bayes-optimal predictive* (1), which knows only the task prior. We call the NLL difference between the closed-form FA-Gaussian of §4 and the Bayes-optimal predictive the *achievable gap*.

An in-context method learns a map  $(\mathcal{D}, x_S) \mapsto q(y)$  by minimizing NLL over tasks drawn from a training distribution [8, 26, 9]. Throughout, models are trained only on queries with *at most one* missing sensor, and evaluated on patterns with up to three missing. Two- and three-sensor deletions are therefore never seen in training.

## 3 Scalar experts cannot fuse sensors with shared noise

Consider the linear-Gaussian model  $x_j = a_j y + d_j u + \sigma_j \epsilon_j$  with  $y, u, \epsilon_j \sim \mathcal{N}(0, 1)$ , where  $u$  is a nuisance shared by all sensors. A product-of-experts (PoE) predictor gives each sensor a factor

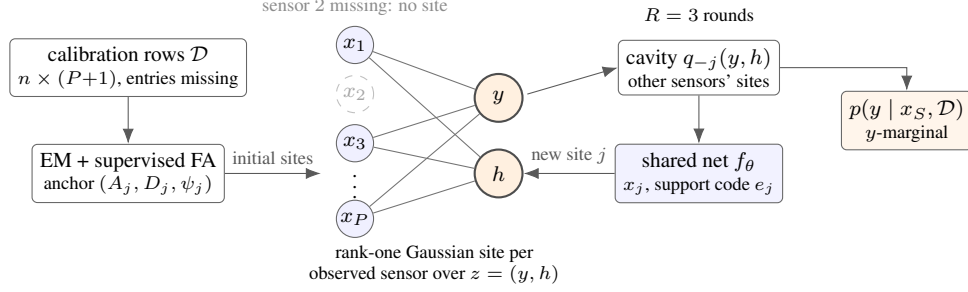


Figure 1: Lifted cavity network. A supervised factor analysis estimated from the incomplete calibration rows gives every sensor a rank-one Gaussian site over the target  $y$  and a shared nuisance  $h$ . Missing sensors contribute no site, so any subset and any number of sensors are handled by omission. For  $R$  rounds, a shared network replaces each observed sensor’s site using its cavity (the posterior from the other sensors) and an encoding of that sensor’s calibration scatter. The prediction is the  $y$ -marginal of the product.

84  $t_j(y; x_j)$  and predicts  $q(y | x_S) \propto p(y) \prod_{j \in S} t_j(y; x_j)$ . Omitting missing sensors is what makes the  
 85 construction mask-consistent.

86 **Proposition 1** (Scalar experts). *Let each  $t_j$  be Gaussian in  $y$  with positive precision, depend on  $x_j$ ,  
 87 and not depend on  $S$ . Suppose  $q$  equals the Bayes posterior for every single-sensor query  $S = \{j\}$ .  
 88 Then for a pair  $\{j, k\}$  with  $a_j a_k \neq 0$ ,  $q$  equals the Bayes posterior if and only if  $d_j d_k = 0$ . Moreover,  
 89  $q$ ’s pair posterior mean gives  $x_j$  a coefficient with the sign of  $a_j$ , whereas the Bayes coefficient can  
 90 have the opposite sign.*

91 The proof (Appendix A) shows that matching single-sensor queries forces  $t_j(y; x_j) \propto p(x_j | y)$ , so  $q$   
 92 is the naive-Bayes posterior. For example, take  $a = (1, 1)$ ,  $d = (1, 0.5)$  and  $\sigma = (0.05, 0.05)$ . The  
 93 Bayes coefficient of  $x_1$  is  $+0.50$  when  $x_1$  is alone and  $-0.96$  in the pair: the second sensor lets the  
 94 predictor estimate  $u$  and subtract it from the first.

95 EP [24] refines each factor using its *cavity*, the approximate posterior given the other sensors, and  
 96 learned EP amortizes the refinement [12, 6, 16]. With a Gaussian over  $y$  alone, however, the cavity  
 97 summarizes the other sensors by a scalar mean and variance, discarding what they reveal about  $u$ .  
 98 Proposition 1 does not exclude every scalar-cavity network, but the failure is visible in practice (§6.4).

## 99 4 Lifted cavity networks

100 **Lifted sites.** Let  $z = (y, h) \in \mathbb{R}^{1+K}$ , where  $h$  is a  $K$ -dimensional nuisance with prior  $\mathcal{N}(0, I)$   
 101 and  $y$  has prior  $\mathcal{N}(\mu_y, v_y)$ . Write the prior in natural parameters  $(\Lambda_0, \eta_0)$ . Each observed sensor  
 102 contributes a rank-one Gaussian site, which corresponds to a linear-Gaussian measurement  $\tilde{x}_j =$   
 103  $w_j^\top z + e_j$  with  $e_j \sim \mathcal{N}(0, \psi_j)$ :

$$\Lambda_j = \frac{w_j w_j^\top}{\psi_j}, \quad \eta_j = \frac{w_j o_j}{\psi_j}. \quad (2)$$

104 The prediction is the  $y$ -marginal of

$$q(z | x_S) = \mathcal{N}(z; \Lambda^{-1} \eta, \Lambda^{-1}), \quad \Lambda = \Lambda_0 + \sum_{j \in S} \Lambda_j, \quad \eta = \eta_0 + \sum_{j \in S} \eta_j. \quad (3)$$

105 **Proposition 2** (Exactness). *Suppose  $x_j = A_j y + D_j^\top h + c_j + e_j$ , with independent  $e_j \sim \mathcal{N}(0, \psi_j)$   
 106 and the priors above. Set  $w_j = (A_j, D_j)$  and  $o_j = x_j - c_j$ . Then (3) equals  $p(z | x_S)$  for every  
 107  $S \subseteq \{1, \dots, P\}$ , and its  $y$ -marginal is the Bayes predictive for every missingness pattern.*

108 Shared noise of rank  $K$  is absorbed by  $K$  nuisance dimensions, so the obstruction of Proposition 1  
 109 disappears: the lifted product differences out  $h$  automatically. Fewer nuisance dimensions cannot  
 110 suffice, so the lifting dimension must match the rank of the shared noise:

111 **Proposition 3** (Lifting dimension). *Let  $x = ay + Du + e$  with  $y \sim \mathcal{N}(0, 1)$ ,  $u \sim \mathcal{N}(0, I_K)$ ,  
 112  $D \in \mathbb{R}^{P \times K}$ , independent Gaussian  $e$ , and every  $a_j \neq 0$ . Suppose that after deleting any row,*

113  $L = [a, D]$  still contains two disjoint nonsingular  $(1 + K) \times (1 + K)$  submatrices, which holds  
 114 for almost every  $L$  once  $P \geq 2K + 3$ . If a product of mask-independent sites of the form (2) with  
 115  $o_j = x_j - c_j$ , over  $z = (y, h)$  with  $h \in \mathbb{R}^{K'}$  and a Gaussian prior, reproduces  $p(y \mid x_S)$  for every  $S$   
 116 with  $|\bar{S}| \leq 2$ , then  $K' \geq K$ .

117 The proof (Appendix A) shows that the conditionals on single sensors and pairs identify the covariance  
 118 of the sensors, which a product of such sites can only represent as a matrix of rank  $1 + K'$  plus a  
 119 diagonal; the identifiability argument of factor analysis [1] excludes this when  $K' < K$ . Our main  
 120 synthetic panels ( $K = 1$ ,  $P = 5$ ) meet the condition, so for products of static sites one nuisance  
 121 dimension is necessary and sufficient.

122 **Closed-form initialization from the support.** Before any learning, we estimate the factor model  
 123 from  $\mathcal{D}$  alone. EM for a Gaussian with missing entries [4, 21] gives the joint mean and covariance  
 124 of  $(x, y)$  from the incomplete calibration rows. Then  $A = \text{Cov}(x, y)/\text{Var}(y)$ , and principal-factor  
 125 iterations on the residual covariance  $\text{Cov}(x) - AA^\top \text{Var}(y)$  give  $(D, \psi)$  [31, 32]. With sensors  
 126 standardized by their support statistics,  $o_j = \tilde{x}_j + A_j \mu_y$ . We call the resulting predictor *FA-Gaussian*;  
 127 it is exact for every mask under the fitted model.

128 **Learned cavity refinement.** Real sensors are nonlinear, real noise is not exactly factor-structured,  
 129 and  $n = 48$  calibration rows make plug-in estimates noisy. We therefore refine the sites for  
 130  $R$  rounds. In each round, every observed sensor  $j$  removes its site from (3) to form the cavity  
 131  $q_{-j}(z) = \mathcal{N}(m_{-j}, V_{-j})$ , the posterior over target and nuisance given the other sensors. The cavity  
 132 predicts the reading as  $\hat{x}_j = A_j(m_{-j,y} - \mu_y) + D_j^\top m_{-j,h}$ , with variance  $v_j = w_j^\top V_{-j} w_j + \psi_j$ .  
 133 A shared two-layer network  $f_\theta$  reads the reading  $\hat{x}_j$ , an encoding  $e_j$  of sensor  $j$ 's calibration  
 134 scatter, the anchor  $(A_j, D_j, \log \psi_j)$ , the cavity  $(m_{-j}, \log V_{-j,yy}, V_{-j,yh})$ , the raw and standardized  
 135 residuals of  $\tilde{x}_j$  against  $\hat{x}_j$ ,  $\log v_j$  and the fraction of observed sensors. It outputs a correction  
 136  $(\delta_j^{\text{off}}, \delta_j^{\text{slope}}, \delta_j^D, s_j)$ , and the site is replaced by the local linearization of the sensor's response at the  
 137 cavity mean  $\bar{y} = m_{-j,y}$ :

$$\begin{aligned} w_j &= (A_j + \delta_j^{\text{slope}}, D_j + \delta_j^D), & o_j &= \tilde{x}_j - g_j(\bar{y}) + w_{j,y} \bar{y}, \\ g_j(\bar{y}) &= A_j(\bar{y} - \mu_y) + \delta_j^{\text{off}}, & \psi_j &\leftarrow \psi_j e^{4 \tanh(s_j/4)}. \end{aligned} \quad (4)$$

138 The encoding  $e_j \in \mathbb{R}^{16}$  is a DeepSets network [36] over sensor  $j$ 's calibration scatter, so  $f_\theta$  can  
 139 recognize a sensor's nonlinearity from the support. The last layer of  $f_\theta$  is initialized at zero.

140 If  $f_\theta$  were replaced by the statistical linear regression of a *known* response function under the cavity,  
 141 these rounds would be EP with posterior linearization [3, 7]. The network amortizes the unknown  
 142 response from the calibration rows, in the spirit of algorithm unrolling [11, 25].

143 **Proposition 4** (Structural guarantees). *For any parameters  $\theta$ :*

- 144 (i) *the prediction does not depend on the readings of missing sensors;*
- 145 (ii) *permuting sensors together with their calibration columns leaves the prediction unchanged,*  
 146 *and the same  $\theta$  applies for every  $P$ ;*
- 147 (iii) *when the last layer of  $f_\theta$  is zero, the network equals the FA-Gaussian closed form of its*  
 148 *anchor, so the minimal training risk over  $\theta$  is at most that closed form's.*

149 **Training.** We minimize the predictive NLL over episodes drawn from a pool of 60,000 synthetic  
 150 tasks (128 tasks  $\times$  8 queries per step, 20,000 AdamW steps with cosine decay, query masks with at  
 151 most one missing sensor). We use  $K = 1$  and  $R = 3$ . The model has about 8,000 parameters and  
 152 trains in 15–20 minutes on one CPU core, 14 times less than the transformer baseline (Table 6). The  
 153 same recipe trains every learned baseline except the transformer, which gets fresh tasks at every step.

## 154 5 Related work

155 **Products of experts** handle missing modalities by omission [13, 35, 22]; our negative result says  
 156 when experts over the target alone are insufficient. **Learned message passing**, through learned  
 157 EP messages [12, 6, 16] or unrolled inference [11, 25], amortizes inference inside a fixed model;  
 158 here the model is unknown and must be read from a small calibration set, and the cavity is over

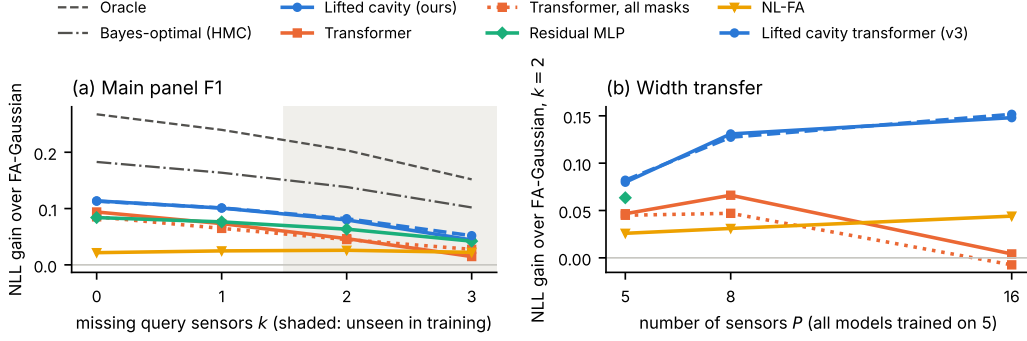


Figure 2: NLL gain over the closed-form FA-Gaussian anchor (higher is better). (a) Main panel F1 by number of missing query sensors; shaded patterns were never seen in training. Grey lines are the privileged oracle and the Bayes-optimal predictive. (b) Width transfer with two missing sensors: every learned model was trained on five sensors, and the residual MLP exists only at  $P = 5$ . Dashed blue: the lifted cavity transformer of §6.6, descriptive on these panels.

159 a lifted latent. **Supervised learning with missing values** [30, 21, 17] studies one task at a time;  
 160 NeuMiss [20] unrolls the Bayes predictor of a Gaussian model but is trained per dataset and keyed to  
 161 feature identities. **In-context prediction** with neural processes [8, 19] and prior-fitted networks such  
 162 as TabPFN and TabICL [26, 14, 15, 28] uses generic set or transformer architectures; we compare  
 163 against such a transformer trained on our task distribution and test lifted sites as its output layer.  
 164 **Fusion under unknown correlations** [18, 34, 5, 29] is learned or bounded per system, with fixed  
 165 sensor sets.

## 166 6 Experiments

### 167 6.1 Setup

168 **Synthetic tasks.** Each task draws, for every sensor  $j$ , a response  $x_j = a_j y + b_j + 0.4 \sin(f_j y) +$   
 169  $d_j u + \sigma_j \epsilon_j$  with  $a_j = \pm \mathcal{U}[0.6, 1.8]$  (random sign),  $b_j \sim \mathcal{N}(0, 0.4^2)$ ,  $f_j \sim \mathcal{U}[0.5, 2.5]$ , shared-noise  
 170 loading  $d_j \sim \mathcal{N}(0, 0.7^2)$ ,  $\sigma_j \sim \mathcal{U}[0.15, 0.8]$  and  $y, u, \epsilon_j \sim \mathcal{N}(0, 1)$ . The support has  $n = 48$  rows  
 171 with 20% of sensor entries missing at random, and 48 query rows are scored under every mask with  
 172  $k$  missing sensors (or 20 seeded masks when there are more). The main panel (F1) has 256 fresh  
 173 tasks with five sensors. Six further panels of 128 tasks change one factor at a time: 8 or 16 sensors,  
 174 nonlinearity 0.8 or 0, and 16 or 96 support rows.

175 **Methods.** *Support-only closed forms:* FA-Gaussian, EM-Gaussian, NL-FA (cubic responses with a  
 176 tuned curvature penalty and exact grid inference, a strong plug-in nonlinear model), evidence-tuned  
 177 Bayesian linear regression (BLR), ridge, and, post hoc, a Gaussian process. *Learned in-context*  
 178 *models*, all trained on the same task family and masks: our lifted cavity network (8k parameters, three  
 179 seeds); a residual MLP that receives the same FA anchor and support encodings (15k parameters, five  
 180 sensors only); the scalar-cavity network of §3; and a TabPFN-v2-style cell-token transformer [15]  
 181 trained on fresh tasks at every step (203k parameters, 30,000 steps). A second transformer was trained  
 182 on *all* masks, to separate architecture from training coverage. *Privileged references:* the oracle, and  
 183 the Bayes-optimal predictive estimated by HMC with the shared nuisance marginalized analytically  
 184 (Appendix C).

185 **Protocol.** NLL is the primary metric; coverage, CRPS [10] and RMSE are secondary. Intervals are  
 186 paired 95% intervals over tasks, with per-task NLL averaged over training seeds. Three evaluation  
 187 protocols were committed to version control before their panels existed, and every endpoint is reported  
 188 (Table 5, Appendix D).

Table 1: Main synthetic panel (F1: 256 fresh tasks, five sensors). Mean test NLL by number of missing query sensors; learned models saw at most one missing sensor in training, so  $k = 2, 3$  are unseen patterns. Learned rows average three training seeds (seed s.d.  $\leq 0.002$ ) except the transformers (seed 1 as pre-registered; Table 11 adds a second seed). Last column: share of the achievable gap (FA-Gaussian to Bayes-optimal) closed at  $k = 2$ . Best non-privileged entry per column in bold. <sup>†</sup>Post hoc, scored at  $k = 2$  only. <sup>‡</sup>Designed after protocol v2 (Section 6.6); descriptive on this panel and not bolded.

| Method                                                       | $k=0$         | $k=1$         | $k=2$        | $k=3$        | Gap closed |
|--------------------------------------------------------------|---------------|---------------|--------------|--------------|------------|
| <i>Privileged references</i>                                 |               |               |              |              |            |
| Oracle (true parameters)                                     | −0.419        | −0.250        | −0.026       | 0.287        | −          |
| Bayes-optimal (HMC, true prior)                              | −0.334        | −0.174        | 0.039        | 0.337        | −          |
| <i>Support-only closed forms</i>                             |               |               |              |              |            |
| FA-Gaussian                                                  | −0.152        | −0.011        | 0.177        | 0.439        | 0%         |
| EM-Gaussian                                                  | −0.048        | 0.037         | 0.195        | 0.441        | −13%       |
| NL-FA (cubic, tuned)                                         | −0.173        | −0.036        | 0.151        | 0.417        | 19%        |
| Bayesian linear regression                                   | 0.111         | 0.091         | 0.214        | 0.450        | −26%       |
| Ridge (complete case)                                        | 0.026         | 0.078         | 0.216        | 0.451        | −28%       |
| Ridge (mean-imputed)                                         | 0.198         | 0.280         | 0.399        | 0.582        | −161%      |
| <i>Learned in context (trained on <math>k \leq 1</math>)</i> |               |               |              |              |            |
| <b>Lifted cavity (ours), 8k</b>                              | <b>−0.265</b> | <b>−0.112</b> | <b>0.097</b> | <b>0.395</b> | 58%        |
| Transformer (TabPFN-v2 style), 203k                          | −0.246        | −0.084        | 0.131        | 0.424        | 34%        |
| Residual MLP on FA anchor, 15k                               | −0.236        | −0.087        | 0.114        | 0.397        | 46%        |
| Scalar-cavity network, 3k                                    | 0.055         | 0.178         | 0.336        | 0.555        | −115%      |
| <i>Transformer, trained on all masks</i>                     | −0.237        | −0.076        | 0.132        | 0.411        | 33%        |
| <i>Lifted cavity transformer (v3)<sup>‡</sup>, 203k</i>      | −0.265        | −0.112        | 0.095        | 0.387        | 59%        |

## 6.2 Synthetic tasks with exact references

Table 1 and Figure 2a report the main panel. The lifted cavity network is the best non-privileged method for every number of missing sensors, including the two- and three-sensor deletions it never saw in training. With two missing sensors it closes 58% of the achievable gap between FA-Gaussian and the Bayes-optimal predictive. Two comparators close less: the residual MLP, which receives the same anchor and support encodings with twice the parameters (46%), and the tuned nonlinear plug-in model NL-FA (19%). The TabPFN-v2-style transformer has 25 times the parameters and sees fresh tasks at every step, yet it is worse than the lifted cavity network for every number of missing sensors, by 0.020–0.033 nats (all intervals exclude zero), and closes 34% of the gap. Training it on every mask changes nothing (33%), so its deficit is learning efficiency, not unseen patterns. The residual MLP is the closest competitor: the lifted network beats it by 0.030 nats with all sensors observed and 0.016 with two missing, and the two tie with three missing. EM-Gaussian, BLR and both ridges fall below FA-Gaussian, because they cannot share strength across sensors under missingness. Mean imputation of calibration entries is the worst of all.

## 6.3 Generalization beyond the training distribution

Every learned model was trained on five sensors, nonlinearity 0.4 and 48 calibration rows (Table 2, Figure 2b).

**Width.** The lifted cavity network runs unchanged on 8 and 16 sensors, where it improves on FA-Gaussian by +0.131 [+0.097, +0.165] and +0.148 [+0.115, +0.181] nats with two missing sensors. The residual MLP cannot run on these panels, and EM-Gaussian degrades as the covariance grows. The transformer also runs at any width, but it gains less over FA-Gaussian as sensors are added. On 16 sensors it is no better than the closed form, and the lifted cavity network beats it by +0.144 [+0.127, +0.161] nats. **Support size.** With 16 calibration rows the transformer degrades sharply, while the lifted cavity network keeps a +0.146 [+0.073, +0.219] nat gain over FA-Gaussian; with 96 rows it remains the best method. **Stronger nonlinearity** (0.8 instead of 0.4) is the one shift where the transformer competes. It ties the lifted cavity network with two missing sensors (difference +0.000 [−0.011, +0.012]) and is better by 0.041 nats when all sensors are observed: under strong response nonlinearity alone, Gaussian re-linearization offers no advantage over a generic transformer. **Linear**

Table 2: Generalization beyond the training distribution (fresh panels of 128 tasks). Mean test NLL with two missing query sensors. Every learned model was trained on  $P=5$  sensors, nonlinearity 0.4 and  $n=48$  support rows; the residual MLP is defined only for five sensors. Best non-privileged entry per column in bold; italic rows are controls or descriptive (the lifted cavity transformer was designed after protocol v2) and are not bolded.

| Method                                | $P=8$         | $P=16$        | nonlin. 0.8  | linear       | $n=16$       | $n=96$       |
|---------------------------------------|---------------|---------------|--------------|--------------|--------------|--------------|
| Oracle                                | -0.598        | -1.142        | -0.086       | -0.066       | -0.060       | -0.043       |
| FA-Gaussian                           | -0.296        | -0.797        | 0.334        | <b>0.009</b> | 0.510        | 0.114        |
| EM-Gaussian                           | -0.160        | 2.977         | 0.311        | 0.034        | 1.140        | 0.112        |
| NL-FA                                 | -0.327        | -0.841        | 0.258        | 0.030        | 0.496        | 0.073        |
| BLR                                   | 0.615         | 0.058         | 0.331        | 0.061        | 3.690        | 0.125        |
| Lifted cavity (ours)                  | <b>-0.427</b> | <b>-0.945</b> | <b>0.191</b> | 0.031        | <b>0.363</b> | <b>0.052</b> |
| Transformer                           | -0.362        | -0.801        | <b>0.191</b> | 0.085        | 0.523        | 0.081        |
| Residual MLP                          | -             | -             | 0.235        | 0.036        | 0.371        | 0.074        |
| <i>Transformer, all masks</i>         | -0.343        | -0.790        | 0.190        | 0.080        | 0.483        | 0.080        |
| <i>Lifted cavity transformer (v3)</i> | -0.423        | -0.949        | 0.179        | 0.029        | 0.457        | 0.047        |

Table 3: Real sensor networks: mean test NLL on later, held-out periods. Beijing: 12 target stations  $\times$  weekly episodes in 2016–2017 with natural missingness, plus extra dropped sensors. Air Quality: weekly episodes after October 2004, sensor dropout simulated. Fine-tuning uses only earlier periods and one recipe for every model; lifted-cavity and residual-MLP rows average three seeds. Best entry per column in bold. <sup>†</sup>Post hoc. <sup>‡</sup>Designed after protocol v2 (Section 6.6); post hoc here and not bolded.

| Method                                                   | Beijing PM <sub>2.5</sub> (11 stations) |              |              | Air Quality CO |               | Air Quality NO <sub>2</sub> |               |
|----------------------------------------------------------|-----------------------------------------|--------------|--------------|----------------|---------------|-----------------------------|---------------|
|                                                          | natural                                 | +3           | +6           | $k=0$          | $k=2$         | $k=0$                       | $k=2$         |
| EM-Gaussian                                              | 1.032                                   | 0.742        | 0.569        | 0.074          | 0.085         | -0.237                      | -0.175        |
| FA-Gaussian                                              | 0.677                                   | 0.615        | 0.562        | 0.052          | 0.116         | -0.196                      | -0.155        |
| NL-FA                                                    | 0.938                                   | 0.744        | 0.610        | 0.144          | 0.091         | -0.090                      | -0.142        |
| Bayesian linear regression                               | 0.399                                   | 0.406        | 0.442        | 1.037          | 0.216         | -0.055                      | -0.178        |
| Ridge (complete case)                                    | 0.425                                   | 0.432        | 0.458        | 0.824          | 0.208         | -0.200                      | -0.176        |
| GP, complete case <sup>†</sup>                           | -                                       | -            | -            | 3.364          | 0.503         | 25.217                      | -0.198        |
| $k$ -nearest neighbours <sup>†</sup>                     | 0.508                                   | 0.514        | 0.540        | 0.571          | 0.353         | -0.010                      | -0.063        |
| Lifted cavity, zero-shot                                 | 0.826                                   | 0.661        | 0.561        | 0.044          | 0.129         | -0.350                      | -0.233        |
| Transformer, zero-shot                                   | 0.958                                   | 0.743        | 0.687        | -0.017         | 0.081         | -0.360                      | -0.231        |
| <b>Lifted cavity, fine-tuned (ours)</b>                  | 0.205                                   | 0.237        | 0.305        | -0.040         | 0.008         | -0.412                      | -0.337        |
| no synthetic pretraining                                 | 0.247                                   | 0.279        | 0.344        | -0.001         | -0.006        | -0.419                      | -0.316        |
| no cavity input                                          | 0.259                                   | 0.286        | 0.349        | -0.086         | -0.028        | -0.423                      | -0.332        |
| scalar sites ( $K=0$ )                                   | 0.215                                   | 0.252        | 0.347        | -0.026         | -0.003        | -0.359                      | -0.302        |
| Transformer, fine-tuned                                  | <b>0.087</b>                            | <b>0.121</b> | <b>0.236</b> | <b>-0.163</b>  | <b>-0.052</b> | <b>-0.485</b>               | <b>-0.381</b> |
| Residual MLP, fine-tuned                                 | -                                       | -            | -            | 0.025          | 0.110         | -0.246                      | -0.147        |
| <i>Lifted cavity transformer, fine-tuned<sup>‡</sup></i> | 0.122                                   | 0.160        | 0.259        | -              | -             | -                           | -             |

217 **sensors.** When the true sensors are linear, FA-Gaussian is nearly the right model and is best; the  
218 learned re-linearization then costs 0.022 nats with two missing sensors, and the transformer 0.076.

#### 219 6.4 What the lifting buys

220 Ablations on F1 (Table 8, Appendix F) isolate the components. **Lifting** is the largest factor: scalar  
221 sites ( $K=0$ ) with otherwise identical machinery and training lose 0.134 [+0.120, +0.148] nats with  
222 two missing sensors and end up worse than the untrained FA-Gaussian anchor, which is Proposition 1  
223 in practice. **Cavity conditioning** adds 0.026 [+0.022, +0.029] nats on top of lifted static sites.  
224 A second nuisance dimension ( $K=2$ ) changes nothing, as expected for one-factor noise. The  
225 scalar-cavity network of the earlier design, retrained with the same 60,000-task recipe, remains far  
226 behind FA-Gaussian (Appendix D).

#### 227 6.5 Real sensor networks

228 **Beijing multi-site air quality [37].** The task is to predict log PM<sub>2.5</sub> at each of 12 monitoring  
229 stations from the other 11, using hourly data with natural missingness. Each episode is one station and

Table 4: Protocol v3 (new panels and targets). Mean test NLL. G1: 256 fresh five-sensor tasks by number of missing query sensors  $k$ ; G3: 128 fresh 16-sensor tasks,  $k = 2$ . Beijing: log NO<sub>2</sub> and log CO at each station from the other 11 stations, 2016–2017, natural missingness and six further sensors removed; learned models fine-tuned with the recipe of protocol v2. Best entry per column in bold.

| Method                                 | G1, $P=5$     |               |              |              | G3<br>$P=16$  | Beijing NO <sub>2</sub> |              | Beijing CO   |              |
|----------------------------------------|---------------|---------------|--------------|--------------|---------------|-------------------------|--------------|--------------|--------------|
|                                        | $k=0$         | $k=1$         | $k=2$        | $k=3$        |               | natural                 | +6           | natural      | +6           |
| FA-Gaussian                            | −0.189        | −0.054        | 0.129        | 0.396        | −0.806        | 0.684                   | 0.506        | 0.384        | 0.334        |
| Bayesian linear regression             | 0.013         | 0.022         | 0.153        | 0.401        | 0.063         | 0.275                   | 0.365        | 0.170        | 0.248        |
| Lifted cavity network, 8k              | <b>−0.329</b> | <b>−0.172</b> | 0.040        | 0.348        | −0.934        | 0.282                   | 0.334        | 0.149        | 0.229        |
| Transformer, 203k                      | −0.315        | −0.147        | 0.073        | 0.378        | −0.800        | <b>0.199</b>            | 0.296        | <b>0.019</b> | <b>0.176</b> |
| <b>Lifted cavity transformer, 203k</b> | <b>−0.329</b> | <b>−0.172</b> | <b>0.039</b> | <b>0.340</b> | <b>−0.950</b> | 0.220                   | <b>0.291</b> | 0.054        | 0.187        |

one week, with 48 labeled support hours and 48 query hours. The test period (2016-01 to 2017-02, 718 episodes) was never examined before protocol v2 was committed. Learned models are fine-tuned on episodes from 2013–2015 with one fixed recipe; the protocol’s analysis is cluster-robust over weeks.

With 11 sensors and 48 rows, the joint-covariance closed forms overfit, and evidence-tuned BLR is the strongest closed form. The fine-tuned lifted cavity network improves on it by 0.194 [+0.131, +0.257] nats under natural missingness, and by 0.137 [+0.104, +0.170] with six further sensors removed. The fine-tuned transformer is better still. The lifted cavity network’s gain over it is −0.118 [−0.145, −0.091] nats under natural missingness and −0.069 [−0.080, −0.058] with six sensors removed, so endpoint E4b fails. Before fine-tuning the order is reversed: zero-shot, the lifted cavity network is ahead by 0.127 [+0.057, +0.197] nats with six sensors removed. Synthetic pretraining helps: the same network fine-tuned from initialization is 0.041 nats worse [+0.033, +0.050].

**UCI Air Quality [2].** Five metal-oxide sensors on one device predict the reference analyzers’ log CO and log NO<sub>2</sub>, in weekly episodes after October 2004. These weeks were examined in protocol v1, so this is secondary evidence. With two simulated missing sensors, the fine-tuned lifted cavity network improves on FA-Gaussian by +0.107 [+0.015, +0.200] nats for CO and +0.181 [+0.076, +0.287] for NO<sub>2</sub>. The fine-tuned transformer is again better: the lifted network’s gain over it is −0.060 [−0.106, −0.014] (CO) and −0.044 [−0.074, −0.014] (NO<sub>2</sub>). On CO, the cavity input stops helping after fine-tuning.

**Why the transformer wins on real data.** The advantage is broad: the transformer is better on 89% of Beijing test episodes, by 0.15–0.16 nats in each of the three easier quartiles of closed-form difficulty. The lifted network underfits rather than overfits: the transformer’s final training NLL is 0.19 nats lower, about the test gap. Neither a second nuisance dimension (0.205) nor five times more fine-tuning steps (0.169, against 0.207; Table 10) closes the gap to the transformer (0.087), and per-episode nonparametrics do worse still ( $k$ -nearest neighbours 0.508; a Gaussian process is the worst method on Air Quality). We therefore hypothesized that the limitation lies in how the sites are refined, since each correction sees only its own sensor’s reading, calibration scatter and cavity, rather than in the site family. Protocol v3 tests this.

## 6.6 Lifted sites as a transformer output layer (protocol v3)

Protocol v2 left a split: structure wins on synthetic tasks, the transformer on real networks after fine-tuning. The *lifted cavity transformer* (LCT) keeps the transformer’s backbone, recipe and seed and replaces only its output head: a zero-initialized linear map turns each observed query cell’s final token into the correction ( $\delta_j^{\text{off}}, \delta_j^{\text{slope}}, \delta_j^D, s_j$ ) of that sensor’s site in (4), linearized at the FA-Gaussian posterior mean, and the prediction is the  $y$ -marginal of the product of observed sites. Missing sensors contribute no site and their readings never enter; at initialization the LCT is its anchor’s closed form. Because the hypothesis was formed after the v2 results, v3 uses only new data, two fresh synthetic panels (G1, five sensors; G3, sixteen) and two Beijing pollutants never scored before (NO<sub>2</sub> and CO), and its four endpoints were committed before any of these existed (Table 5).

On the fresh synthetic panels, the lifted output layer helps the transformer (Table 4). With two missing sensors the LCT beats the transformer with the identical backbone, recipe and seed by 0.035 [+0.027, +0.042] nats on G1 and 0.150 [+0.133, +0.167] on sixteen sensors (E5, E6 pass). It

Table 5: Every pre-registered endpoint of protocols v2 and v3, each committed before its panels or targets existed (protocol v1: Appendix D). Gain in nats of NLL, positive when the first-named model is better; paired 95% intervals, cluster-robust over weeks on Beijing. Pass: gain  $\geq 0.01$  and lower bound  $> 0$ , except E7b (lower bound  $> -0.02$ ).

| ID  | Data                      | Comparison                                     | Gain [95% CI]           | Result |
|-----|---------------------------|------------------------------------------------|-------------------------|--------|
| E1  | F1, $k = 2$               | lifted cavity network vs NL-FA                 | +0.054 [+0.043, +0.065] | pass   |
| E2  | F1, $k = 2$               | lifted cavity network vs transformer           | +0.033 [+0.027, +0.040] | pass   |
| E3a | F3 (16 sensors), $k = 2$  | lifted cavity network vs FA-Gaussian           | +0.148 [+0.115, +0.181] | pass   |
| E3b | F3 (16 sensors), $k = 2$  | lifted cavity network vs transformer           | +0.144 [+0.127, +0.161] | pass   |
| E4a | Beijing PM <sub>2.5</sub> | lifted cavity network (FT) vs BLR              | +0.194 [+0.131, +0.257] | pass   |
| E4b | Beijing PM <sub>2.5</sub> | lifted cavity network (FT) vs transformer (FT) | -0.118 [-0.145, -0.091] | fail   |
| E5  | G1, $k = 2$               | LCT vs transformer                             | +0.035 [+0.027, +0.042] | pass   |
| E6  | G3 (16 sensors), $k = 2$  | LCT vs transformer                             | +0.150 [+0.133, +0.167] | pass   |
| E7a | Beijing NO <sub>2</sub>   | LCT (FT) vs lifted cavity network (FT)         | +0.062 [+0.024, +0.100] | pass   |
| E7b | Beijing NO <sub>2</sub>   | LCT (FT) vs transformer (FT), non-inferiority  | -0.021 [-0.093, +0.052] | fail   |

271 matches the small lifted network on five sensors (+0.002 [-0.002, +0.006]) and exceeds it on sixteen  
272 (+0.017 [+0.004, +0.029]), and it learns faster: its training NLL reaches the transformer’s final  
273 value after 10,750 of 30,000 steps (Figure 3). On the protocol-v2 panels (descriptive) it closes 59%  
274 of the achievable gap on F1 (lifted network 58%), but with 16 calibration rows it inherits part of  
275 the transformer’s degradation (Table 2). After fine-tuning on Beijing NO<sub>2</sub>, it beats the fine-tuned  
276 lifted network by +0.062 [+0.024, +0.100] nats (E7a passes), recovering 75% of the gap between  
277 the lifted network (0.282) and the transformer (0.199). Against the transformer the difference is  
278 -0.021 [-0.093, +0.052]; the interval includes zero but extends below the margin of -0.02, so  
279 non-inferiority (E7b) is not established. With six more stations missing, the two tie (+0.005 [-0.027,  
280 +0.037]). The pattern repeats on Beijing CO (descriptive) and on the PM<sub>2.5</sub> target of protocol v2  
281 (post hoc): the LCT recovers 73% and 70% of the gap (Tables 4 and 3).

## 282 6.7 Pre-registered endpoints and calibration

283 Table 5 lists every pre-registered endpoint of protocols v2 and v3. Five of six v2 endpoints pass;  
284 the failure, E4b, is the real-data comparison with the fine-tuned transformer. All five v1 endpoints  
285 passed (Appendix D). Three of the four v3 endpoints pass; the failure, E7b, is non-inferiority to the  
286 fine-tuned transformer on Beijing NO<sub>2</sub>, where the interval is wide. Learned models are close to  
287 nominal 90% coverage (lifted cavity network 0.901 on F1, 0.914 on Beijing), while the closed forms  
288 under-cover (Table 9).

## 289 7 Limitations

290 **Scope.** Exact and Bayes-optimal references exist only for our generator (one-factor noise, sinusoidal  
291 nonlinearity, Gaussian targets). Per-sensor dropout is simulated on Air Quality and mostly simulated  
292 on Beijing, whose natural missingness covers 1.6% of query readings; the gas-drift benchmark  
293 could not separate methods (Appendix D). **Model class.** Gaussian sites cannot represent multimodal  
294 posteriors; we used  $K = 1$  throughout. **Baselines.** The transformers were trained on a CPU budget  
295 (203k parameters, 30,000 steps). Pretrained tabular foundation models (TabPFN v2, TabICL) could  
296 not be run in our sandbox; we provide a script.

## 297 8 Conclusion

298 Products of experts over the target handle missing sensors by construction but provably mishandle  
299 shared noise; Gaussian sites over the target and a nuisance of the noise’s rank are necessary and  
300 sufficient for factor noise. A small lifted cavity network learns fusion more efficiently than a 25-  
301 times-larger transformer trained on the same tasks and extrapolates to unseen missingness patterns  
302 and wider sensor sets, but on real networks it underfits and a fine-tuned transformer is better. As the  
303 transformer’s output layer, lifted sites keep the extrapolation and recover most of that gap; closing the  
304 rest, richer site families and vector targets are natural next steps.

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## 405 A Proofs

406 **Proposition 1.** Under the model,  $p(x_j | y) = \mathcal{N}(x_j; a_j y, c_j)$  with  $c_j = d_j^2 + \sigma_j^2$ .

407 If  $q(y | x_j) = p(y | x_j)$  for all  $x_j$ , then  $p(y) t_j(y; x_j) \propto p(y) p(x_j | y)$  as functions of  $y$ . Hence  
 408  $t_j(y; x_j) = \kappa_j(x_j) p(x_j | y)$ , which is Gaussian in  $y$  with precision  $a_j^2/c_j > 0$ . For a pair,  $q$  is  
 409 therefore the naive-Bayes posterior  $p(y) p(x_j | y) p(x_k | y)$ . Its posterior mean is linear in  $(x_j, x_k)$   
 410 with coefficients

$$(a_j/c_j, a_k/c_k) / \pi_q, \quad \pi_q = 1 + a_j^2/c_j + a_k^2/c_k,$$

411 so the coefficient of  $x_j$  has the sign of  $a_j$ .

412 The Bayes posterior uses  $C = \text{Cov}(x_j, x_k | y)$ , with off-diagonal entry  $d_j d_k$ . Its mean coefficients  
 413 are  $(C^{-1}a)^\top / \pi$ , with  $\pi = 1 + a^\top C^{-1}a$ . Both posteriors are Gaussian, so they coincide for all  
 414  $(x_j, x_k)$  if and only if  $C^{-1}a = \text{diag}(C)^{-1}a$ . Equivalently  $a = C \text{diag}(C)^{-1}a$ , whose off-diagonal  
 415 components are  $d_j d_k a_k / c_k = 0$  and  $d_j d_k a_j / c_j = 0$ . With  $a_j a_k \neq 0$  this holds if and only if  
 416  $d_j d_k = 0$ .

417 For the sign flip, take  $a = (1, 1)$ ,  $d = (1, 0.5)$  and  $\sigma = (0.05, 0.05)$ . Then the Bayes coefficient of  
 418  $x_1$  is  $+0.499$  alone and  $-0.959$  in the pair.  $\square$

419 **Proposition 2.** Given  $z = (y, h)$ , the readings are independent:  $x_j | z \sim \mathcal{N}(w_j^\top z + c_j, \psi_j)$ . Hence

$$p(z | x_S) \propto p(z) \prod_{j \in S} \mathcal{N}(x_j; w_j^\top z + c_j, \psi_j).$$

420 In natural parameters each factor contributes  $w_j w_j^\top / \psi_j$  to the precision and  $w_j(x_j - c_j) / \psi_j$  to the  
 421 shift, which is (2) with  $o_j = x_j - c_j$ . The prior is Gaussian, so the product is the Gaussian (3). This  
 422 holds for every  $S$  because the factors of unobserved sensors are absent from both sides. Marginalizing  
 423  $h$  gives  $p(y | x_S)$ .  $\square$

424 **Proposition 3.** *Step 1: the product is a latent-factor model.* By the computation in the proof of  
 425 Proposition 2 read backwards, a product of sites (2) with  $o_j = x_j - c_j$  is the exact posterior over  $z$  of  
 426 the model  $M_q$ :  $z \sim \mathcal{N}(\mu_z, \Sigma_z)$  and, independently given  $z$ ,  $x_j \sim \mathcal{N}(w_j^\top z + c_j, \psi_j)$ . Its  $y$ -marginal  
 427 is  $M_q$ 's conditional  $p_q(y | x_S)$ , and  $M_q$ 's sensor covariance is  $\Sigma_q = W \Sigma_z W^\top + \text{diag}(\psi)$  with  
 428  $W \in \mathbb{R}^{P \times (1+K')}$ , so  $\Sigma_q - \text{diag}(\psi)$  has rank at most  $1 + K'$ .

429 *Step 2: conditionals on  $|S| \leq 2$  identify the covariance.* Write  $V = \text{Var}(y)$ ,  $c = \text{Cov}(x, y)$  and  
 430  $\Sigma = \text{Cov}(x)$ . The case  $S = \emptyset$  gives  $V$ . For  $S = \{j\}$  the conditional has slope  $\beta_j = c_j / \Sigma_{jj}$   
 431 and variance  $r_j = V - c_j \beta_j$ , and  $\beta_j \neq 0$  because  $c_j = a_j \neq 0$ . Hence  $c_j = (V - r_j) / \beta_j$  and  
 432  $\Sigma_{jj} = c_j / \beta_j$ . For  $S = \{j, k\}$  the slope vector  $b \neq 0$  solves  $\Sigma_{\{jk\}} b = (c_j, c_k)$ , and whichever  
 433 entry of  $b$  is non-zero determines  $\Sigma_{jk}$  from one row. Since  $M_q$  has the same conditionals, the same  
 434 formulas apply to it, so  $\Sigma_q = \Sigma = LL^\top + \text{diag}(\sigma^2)$ .

435 *Step 3: fewer factors cannot reproduce  $\Sigma$ .* Let  $r = 1 + K$ , so  $LL^\top$  has rank  $r$ . Suppose  $\Sigma = G + \Psi'$   
 436 with  $\Psi'$  diagonal and  $\text{rank}(G) \leq r$ . Fix a row  $i$  and two disjoint row sets  $I_1, I_2$  not containing  $i$   
 437 with  $L_{I_1}, L_{I_2}$  nonsingular. In the  $(r+1) \times (r+1)$  submatrix of  $\Sigma$  with rows  $\{i\} \cup I_1$  and columns  
 438  $\{i\} \cup I_2$ , every entry except  $(i, i)$  lies off the diagonal, so it is shared by  $G$  and  $LL^\top$ . Both  $G$  and  
 439  $LL^\top$  have rank at most  $r$ , so the corresponding minors vanish. Expanding along entry  $(i, i)$ , its  
 440 cofactor is  $\det(L_{I_1} L_{I_2}^\top) \neq 0$ , so  $G_{ii} = (LL^\top)_{ii}$ . Hence  $G = LL^\top$ , which has rank  $r$  [1]. Applying  
 441 this to  $G = W \Sigma_z W^\top$  from Step 1 gives  $1 + K' \geq r = 1 + K$ .

442 For almost every  $L$  all  $r \times r$  submatrices are nonsingular, and  $P \geq 2r + 1 = 2K + 3$  leaves two  
 443 disjoint sets of  $r$  rows after deleting any row.  $\square$

444 **Proposition 4.**

445 (i) Sites of missing sensors are multiplied by the mask before entering (3), and the network's  
 446 per-sensor inputs never include another sensor's raw reading, only cavity statistics built from  
 447 observed sites.

- (ii) Every operation is applied per sensor with shared weights. EM and principal factors are equivariant to permutations of the sensor coordinates, and sums over sensors are permutation-invariant. No input encodes a sensor’s index or the total number of sensors, except the fraction of observed sensors, which is permutation invariant.
- (iii) With a zero last layer,  $\delta^{\text{off}} = \delta^{\text{slope}} = \delta^D = s = 0$ . Then  $w_j = (A_j, D_j)$ ,  $\psi_j$  is unchanged and  $o_j = \tilde{x}_j - A_j(\bar{y} - \mu_y) + A_j\bar{y} = \tilde{x}_j + A_j\mu_y$ . These are the initialization sites, so every round reproduces them and the output is the anchor’s FA-Gaussian closed form. The zero point lies in the parameter set, so the infimum of the training risk is at most that closed form’s.  $\square$

## B Architecture and training details

**Lifted cavity network ( $K = 1$ ,  $R = 3$ , 8,004 parameters).**

- **Support encoder.**  $\phi : \mathbb{R}^3 \rightarrow \mathbb{R}^{32} \rightarrow \mathbb{R}^{32}$  with tanh, applied to each observed calibration entry  $(\tilde{y}_i, \tilde{x}_{ij}, \tilde{r}_{ij})$ . The inputs are the standardized target, the centred reading, and the anchor residual scaled by  $\sqrt{\psi_j}$ . The encoder averages over observed rows and then applies  $\rho : \mathbb{R}^{33} \rightarrow \mathbb{R}^{16}$ , where the extra input is the observation rate.
- **Site network.**  $f_\theta : \mathbb{R}^{28} \rightarrow \mathbb{R}^{64} \rightarrow \mathbb{R}^{64} \rightarrow \mathbb{R}^4$  with tanh and a zero-initialized last layer.
- **Anchor.** EM uses 8 iterations with a  $10^{-4}$  ridge. Principal factors use 40 iterations, with uniquenesses floored at  $10^{-3}$ . Everything is computed in double precision from the support only.

Table 6: Compute. Training on one CPU core (core-hours); inference (ms/task) is single-threaded time per task for 480 queries (ten masks  $\times$  48 rows) on F1. Anchors are precomputed in batch and excluded; the closed-form rows include their unbatched NumPy fits.

| Model                         | Parameters | Training steps | Core-hours | ms/task |
|-------------------------------|------------|----------------|------------|---------|
| Lifted cavity network         | 8,004      | 20,000         | 0.22       | 7.2     |
| Residual MLP on FA anchor     | 15,490     | 20,000         | 0.20       | 1.2     |
| Transformer (TabPFN-v2 style) | 202,562    | 30,000         | 3.11       | 42.9    |
| Lifted cavity transformer     | 202,822    | 30,000         | 3.15       | 55.7    |
| FA-Gaussian (closed form)     | —          | —              | —          | 38.3    |
| NL-FA (closed form)           | —          | —              | —          | 34.4    |

**Training recipe (all synthetic-trained models except the transformer).**

- Pool of 60,000 tasks (seed 5150).
- Minibatches of 128 tasks  $\times$  8 queries.
- AdamW with learning rate  $1.8 \times 10^{-3}$ , cosine decay to 10% and weight decay  $10^{-4}$ .
- Gradient clipping at 5 and 20,000 steps.
- Seeds 1–3.

No hyperparameter search was run for any model.

**Controls.**

- **Residual MLP on the same anchor** (15,490 parameters). It receives the FA-Gaussian prediction, the same per-sensor support encodings (concatenated in sensor order), the readings and mask, and a 55-dimensional support summary. It outputs a residual correction to the FA-Gaussian mean and log-variance, and is defined only for five sensors.
- **Lifted sites without cavity input.** The same sites with the cavity statistics removed.
- **1-D sites** ( $K = 0$ ). The same machinery, with sites over  $y$  only.
- **Repository cavity network.** The three-round scalar-cavity architecture of Appendix D, retrained with the shared recipe.

**Transformer baseline (202,562 parameters).** The transformer follows TabPFN v2 [15]:

- every cell is a token, embedded by a linear map of its value or by a learned missing-value token;
- sensors receive random per-task feature embeddings, so the model is width-agnostic and permutation equivariant in distribution;
- the target column has its own embedding, and query targets use a learned token;
- each of 4 blocks alternates attention across the columns of a row with attention across rows within a column (all rows attend to the support rows only), followed by an MLP ( $d = 64$ , 4 heads, hidden width 128, pre-LayerNorm);
- targets are standardized by the support mean and standard deviation, and the output is a Gaussian mean and log-variance.

It was trained on fresh tasks at every step (16 tasks  $\times$  16 queries, AdamW with learning rate  $10^{-3}$ , 1,000 warm-up steps, cosine decay, clipping at 1, 30,000 steps), with the same query masks as the other models. A second copy (*all masks*) saw query masks with 0–4 missing sensors, drawn uniformly. Evaluation uses a fixed seed for the random feature embeddings.

**Lifted cavity transformer (protocol v3, 202,822 parameters).** The backbone is the transformer above, unchanged. Its output head is replaced by a zero-initialized linear map from each query cell’s final token to four numbers ( $\delta_j^{\text{off}}, \delta_j^{\text{slope}}, \delta_j^D, s_j$ ), used exactly as in (4) with the linearization point  $\bar{y}$  set to the anchor’s closed-form posterior mean (one round,  $K = 1$ , gradients not propagated through  $\bar{y}$ ). Readings of missing sensors are replaced by the missing-value token before the backbone and contribute no site, so they never enter the prediction (a unit test checks both this and the identity with the anchor’s closed form at initialization). Anchors are computed on the fly for each training task. Training, masks, seed and evaluation are those of the transformer. Figure 3 compares the two training curves.

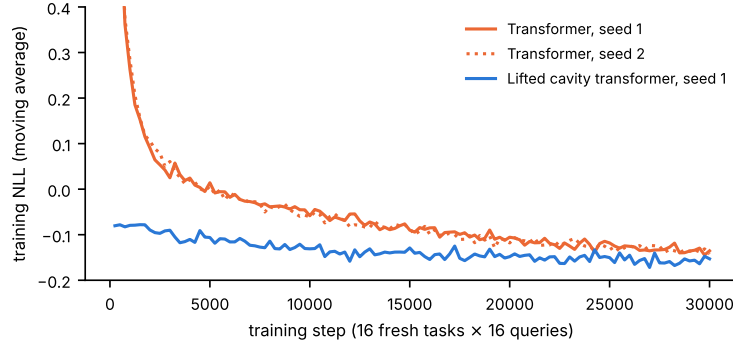


Figure 3: Training NLL (moving average; fresh tasks with at most one missing sensor) of the transformer and the lifted cavity transformer, which share backbone, recipe and seed. The lifted output layer starts at its anchor’s closed form, so it begins far lower; the plain transformer must first learn fusion from scratch.

## C Baselines and references

### Support-only closed forms.

- **EM-Gaussian:** the conditional of  $y$  given  $x_S$  under the EM joint Gaussian (40 iterations).
- **FA-Gaussian:** the closed form of §4, with EM run for 40 iterations. The networks’ anchors use 8 iterations and no variance inflation, so the reported FA-Gaussian is a slightly stronger baseline than the untrained networks (on F1 with two missing sensors, 0.177 against 0.187).
- **NL-FA:** per-sensor cubic response curves in the standardized target. The intercept and slope are unpenalized; the curvature terms are ridge-penalized ( $\lambda = 100$ ) and clamped to the support’s target range for linear extrapolation. The residual covariance has a one-factor structure. Exact inference is done on a 401-point grid. The degree and  $\lambda$  were tuned on

the development panel only: free degree selection by leave-one-out produced catastrophic extrapolation.

- **BLR**: evidence-maximized Bayesian linear regression [23] on complete-case rows, with predictive variance including parameter uncertainty.
- **Ridge**: complete-case ridge with leave-one-out residual variance, and the original mean-imputed ridge.
- **Gaussian process (post hoc)**: for each episode and observed-sensor set, an ARD squared-exponential kernel plus white noise on the complete-case calibration rows, with hyperparameters by type-II maximum likelihood (one restart).
- **$k$ -nearest neighbours (post hoc)**: on the standardized observed sensors of complete-case calibration rows, with  $k \in \{1, 2, 3, 5, 7, 10\}$  chosen per episode and sensor set by leave-one-out NLL on the support and the leave-one-out mean squared error as predictive variance.

EM-Gaussian, FA-Gaussian and ridge inflate their predictive variance by  $n/(n - |S| - 1)$ .

**Bayes-optimal ceiling.** The ceiling is HMC [27] over the 25 (or 5P) task parameters. It uses the true priors, transformed to unconstrained coordinates. The shared nuisance is marginalized analytically through a rank-one-plus-diagonal covariance, using only the observed entries of each calibration row. The sign of each  $a_j$  is fixed by the support correlation, which is unambiguous because  $|a_j| \geq 0.6$ .

The sampler runs 4 chains per task, each started from a MAP fit. Warm-up has two phases: step-size adaptation (target acceptance 0.75) and diagonal mass-matrix estimation from 150 iterations, followed by 150 more adaptation iterations. We keep 50 draws per chain, thinned by 4, with 12 leapfrog steps.

The predictive density of a query uses the samples exactly, including the query’s own reweighting:  $p(y | x_S, \mathcal{D}) = \sum_m p(y, x_S | \theta_m) / \sum_m p(x_S | \theta_m)$ , with  $p(x_S | \theta_m)$  by trapezoid quadrature on 301 points. On every panel where it was run, R-hat on identifiable parameters was below 1.1 for at least 99.9% of parameters. Run with the true parameters, the predictive formula reproduces the exact oracle to within  $2 \times 10^{-3}$  nats (a unit test).

## D Protocols and audit trail

**How the method arose.** The lifted cavity network replaced an earlier design that failed. That design was a three-round EP-style network whose sites live on  $y$  alone and are updated from the scalar cavity of  $y$ . It was evaluated under pre-registered gates on the same synthetic family. Its first version collapsed to the prior during training. A repaired version, with sites initialized from support moments, trained successfully but was beaten by a ridge regression. On that panel the oracle, the ridge control and the best repaired seed scored:

| Method                                     | NLL, two sensors missing |
|--------------------------------------------|--------------------------|
| Oracle                                     | −0.046                   |
| Ridge control                              | 0.396                    |
| Repaired scalar-cavity network (best seed) | 0.403                    |

Computing the oracle exposed three problems. First, the control was weak: it mean-imputed missing calibration entries, and a support-only EM or factor-analysis Gaussian scores 0.168 or 0.156. Second, scalar sites are structurally limited (§3). Third, the learners were starved: 512 reused tasks and 3,000 updates. Retraining the scalar-cavity architecture with our recipe (60,000 tasks, 20,000 steps) improves it by only about 0.05 nats, and on F1 it remains 0.159 nats worse than the untrained FA-Gaussian (§6.4).

**Protocol v1.** After development on panels that are now retired, five endpoints were committed to version control before their panels were generated:

- fresh synthetic tasks, against FA-Gaussian and against the residual MLP;
- 8-sensor transfer;
- stronger nonlinearity;
- a new real target (NO<sub>2</sub> on the Air Quality device).

563 All five passed (Table 7). Each endpoint was evaluated once, and no model was retrained after the  
564 commit.

Table 7: Protocol v1 endpoints (two missing sensors; positive gain favours the lifted cavity network; pass requires gain  $\geq 0.01$  and a positive lower bound).

| Endpoint | Comparison                                                   | Gain [95% CI]           | Result |
|----------|--------------------------------------------------------------|-------------------------|--------|
| E1       | Fresh tasks, vs FA-Gaussian                                  | +0.074 [+0.064, +0.084] | pass   |
| E2       | Fresh tasks, vs residual MLP                                 | +0.016 [+0.012, +0.020] | pass   |
| E3       | 8 sensors, vs FA-Gaussian                                    | +0.115 [+0.097, +0.132] | pass   |
| E4       | Nonlinearity 0.8, vs residual MLP                            | +0.028 [+0.020, +0.036] | pass   |
| R1       | Air Quality NO <sub>2</sub> , 23 later weeks, vs EM-Gaussian | +0.112 [+0.050, +0.173] | pass   |

565 **Protocol v2.** Protocol v2 was likewise committed before any of its panels or test splits existed. It  
566 adds:

- 567 • the transformer baseline, NL-FA, BLR and the HMC ceiling;
- 568 • wider panels (8 and 16 sensors), and shifts in nonlinearity and support size;
- 569 • the Beijing benchmark, whose test period had never been scored.

570 NL-FA’s settings and BLR’s choice as the strongest real-data closed form were fixed on development  
571 data (Appendix C). The gas-sensor-drift dataset [33] was examined during development and then  
572 excluded, before any test split was built. Its heavy-tailed sensor outliers made the mean NLL of every  
573 Gaussian method dominated by a few episodes, with mean and median differing by more than 100  
574 nats, so it cannot separate methods. One deviation was recorded before any Beijing score existed:  
575 some query hours have no other station observed, and complete-case BLR cannot fit zero features, so  
576 for an empty sensor set it returns the support mean with variance  $s^2(1 + 1/n)$ , like the other closed  
577 forms’ prior predictive.

578 **Protocol v3.** After protocol v2’s real-data endpoint failed, protocol v3 was committed before any  
579 of its panels or targets existed. It fixes one model (the lifted cavity transformer of §6.6, trained  
580 once with the transformer’s recipe and seed and evaluated at its final checkpoint), two new synthetic  
581 panels (G1, G3), two Beijing pollutants never scored by any model (NO<sub>2</sub>, primary; CO, descriptive),  
582 the fine-tuning recipe of protocol v2, and four endpoints (Table 5). The lifted cavity transformer’s  
583 scores on the v2 panels are descriptive only (Appendix F). Two container restarts interrupted its  
584 training before any v3 score existed; it was retrained from scratch with the same seed, and later runs  
585 checkpoint every random state so that a resumed run is bitwise identical to an uninterrupted one (a  
586 unit test). The protocol document logs both events.

587 **Post-hoc analyses.** Everything not named in a protocol is post hoc and labeled as such: the per-  
588 episode Gaussian process and  $k$ -nearest neighbours, the second training seeds of both transformers,  
589 the fine-tuning diagnostics of §6.5, and the exploratory runs of Appendix F.

## 590 E Broader impact

591 Calibrating low-cost environmental sensors widens access to air-quality monitoring. Overconfident  
592 predictions could mislead exposure decisions, which is why we report coverage alongside likelihood.  
593 We recommend validating calibration on each new deployment before relying on predicted intervals.

## 594 F Additional results

Table 8: Ablations on F1 (mean test NLL). Same training recipe for every learned row.

| Variant                                  | $k=0$  | $k=1$  | $k=2$ | $k=3$ |
|------------------------------------------|--------|--------|-------|-------|
| Lifted cavity, $K = 1$ (ours)            | -0.265 | -0.112 | 0.097 | 0.395 |
| Lifted cavity, $K = 2$                   | -0.267 | -0.112 | 0.098 | 0.397 |
| Lifted sites, no cavity input            | -0.221 | -0.074 | 0.123 | 0.399 |
| Scalar sites ( $K = 0$ ), same machinery | -0.070 | 0.061  | 0.232 | 0.473 |
| Scalar-cavity network (retrained)        | 0.055  | 0.178  | 0.336 | 0.555 |
| FA-Gaussian (untrained)                  | -0.152 | -0.011 | 0.177 | 0.439 |

Table 9: Calibration and point accuracy. Coverage of the central 90% predictive interval (target 0.90), continuous ranked probability score (CRPS, lower is better) and RMSE. F1 with two missing sensors; Beijing test with natural missingness (fine-tuned learned models).

| Method               | F1, $k = 2$ |       |       | Beijing, natural |       |       |
|----------------------|-------------|-------|-------|------------------|-------|-------|
|                      | Cov. 90     | CRPS  | RMSE  | Cov. 90          | CRPS  | RMSE  |
| FA-Gaussian          | 0.873       | 0.168 | 0.316 | 0.824            | 0.189 | 0.379 |
| EM-Gaussian          | 0.860       | 0.170 | 0.318 | 0.807            | 0.207 | 0.423 |
| BLR                  | 0.866       | 0.174 | 0.327 | 0.861            | 0.182 | 0.374 |
| Lifted cavity (ours) | 0.901       | 0.160 | 0.305 | 0.914            | 0.171 | 0.357 |
| Transformer          | 0.902       | 0.165 | 0.314 | 0.914            | 0.158 | 0.341 |
| Residual MLP         | 0.902       | 0.162 | 0.309 | –                | –     | –     |

Table 10: Post-hoc fine-tuning variants on Beijing  $\text{PM}_{2.5}$  (test period of protocol v2; mean test NLL). Neither more nuisance dimensions nor longer fine-tuning closes the gap to the fine-tuned transformer.

| Fine-tuned model                                | natural | +3    | +6    |
|-------------------------------------------------|---------|-------|-------|
| Lifted cavity network, standard recipe (seed 1) | 0.207   | 0.237 | 0.304 |
| with $K = 2$                                    | 0.205   | 0.239 | 0.300 |
| with $5 \times$ fine-tuning steps               | 0.169   | 0.201 | 0.276 |
| Transformer, standard recipe                    | 0.087   | 0.121 | 0.236 |

Table 11: Second training seeds of the transformer and the lifted cavity transformer (post hoc). Endpoint comparisons recomputed with each seed and with both averaged; the lifted cavity network always averages its three seeds. Gains in nats, paired 95% intervals (cluster-robust over weeks on Beijing).

| ID  | Data                  | Comparison                    | Seed 1 (protocol)       | Seed 2                  | Seeds averaged          |
|-----|-----------------------|-------------------------------|-------------------------|-------------------------|-------------------------|
| E2  | F1, $k = 2$           | lifted network vs transformer | +0.033 [+0.027, +0.040] | +0.037 [+0.030, +0.044] | +0.035 [+0.029, +0.042] |
| E3b | F3, $k = 2$           | lifted network vs transformer | +0.144 [+0.127, +0.161] | +0.155 [+0.139, +0.171] | +0.149 [+0.134, +0.165] |
| E5  | G1, $k = 2$           | LCT vs transformer            | +0.035 [+0.027, +0.042] | –                       | –                       |
| E6  | G3, $k = 2$           | LCT vs transformer            | +0.150 [+0.133, +0.167] | –                       | –                       |
| E7a | Beijing $\text{NO}_2$ | LCT vs lifted network (FT)    | +0.062 [+0.024, +0.100] | –                       | –                       |
| E7b | Beijing $\text{NO}_2$ | LCT vs transformer (FT)       | -0.021 [-0.093, +0.052] | –                       | –                       |

Table 12: Panel F2 (8 sensors; 128 tasks): mean test NLL by number of missing sensors.

| Method                         | $k=0$  | $k=1$  | $k=2$  | $k=3$  |
|--------------------------------|--------|--------|--------|--------|
| Oracle                         | -0.807 | -0.712 | -0.598 | -0.459 |
| FA-Gaussian                    | -0.462 | -0.387 | -0.296 | -0.183 |
| EM-Gaussian                    | 0.032  | -0.133 | -0.160 | -0.120 |
| NL-FA                          | -0.485 | -0.414 | -0.327 | -0.215 |
| BLR                            | 0.187  | 0.398  | 0.615  | 0.028  |
| Ridge (complete case)          | 0.064  | -0.003 | -0.033 | -0.015 |
| Lifted cavity (ours)           | -0.615 | -0.530 | -0.427 | -0.302 |
| Lifted cavity, $K = 2$         | -0.618 | -0.531 | -0.427 | -0.302 |
| Lifted sites, no cavity        | -0.551 | -0.469 | -0.369 | -0.250 |
| Scalar sites ( $K = 0$ )       | -0.295 | -0.228 | -0.148 | -0.057 |
| Transformer                    | -0.568 | -0.474 | -0.362 | -0.229 |
| Transformer, all masks         | -0.545 | -0.452 | -0.343 | -0.212 |
| Lifted cavity transformer (v3) | -0.611 | -0.526 | -0.423 | -0.299 |

Table 13: Panel F3 (16 sensors; 128 tasks): mean test NLL by number of missing sensors.

| Method                         | $k=0$  | $k=1$  | $k=2$  | $k=3$  |
|--------------------------------|--------|--------|--------|--------|
| Oracle                         | -1.223 | -1.184 | -1.142 | -1.096 |
| FA-Gaussian                    | -0.864 | -0.832 | -0.797 | -0.758 |
| EM-Gaussian                    | 6.300  | 4.851  | 2.977  | 1.553  |
| NL-FA                          | -0.910 | -0.876 | -0.841 | -0.803 |
| BLR                            | 0.095  | 0.078  | 0.058  | 0.054  |
| Ridge (complete case)          | 0.093  | 0.089  | 0.081  | 0.086  |
| Lifted cavity (ours)           | -1.018 | -0.982 | -0.945 | -0.902 |
| Lifted cavity, $K = 2$         | -1.017 | -0.981 | -0.944 | -0.901 |
| Lifted sites, no cavity        | -0.952 | -0.916 | -0.880 | -0.836 |
| Scalar sites ( $K = 0$ )       | -0.658 | -0.627 | -0.591 | -0.554 |
| Transformer                    | -0.886 | -0.846 | -0.801 | -0.747 |
| Transformer, all masks         | -0.869 | -0.832 | -0.790 | -0.738 |
| Lifted cavity transformer (v3) | -1.020 | -0.985 | -0.949 | -0.905 |

Table 14: Panel F4 (nonlinearity 0.8; 128 tasks): mean test NLL by number of missing sensors.

| Method                         | $k=0$  | $k=1$  | $k=2$  | $k=3$ |
|--------------------------------|--------|--------|--------|-------|
| Oracle                         | -0.513 | -0.331 | -0.086 | 0.261 |
| FA-Gaussian                    | 0.050  | 0.170  | 0.334  | 0.568 |
| EM-Gaussian                    | 0.050  | 0.145  | 0.311  | 0.559 |
| NL-FA                          | -0.032 | 0.089  | 0.258  | 0.507 |
| BLR                            | 0.208  | 0.188  | 0.331  | 0.568 |
| Ridge (complete case)          | 0.138  | 0.208  | 0.338  | 0.571 |
| Lifted cavity (ours)           | -0.185 | -0.026 | 0.191  | 0.498 |
| Lifted cavity, $K = 2$         | -0.200 | -0.035 | 0.186  | 0.497 |
| Lifted sites, no cavity        | -0.068 | 0.075  | 0.261  | 0.519 |
| Scalar sites ( $K = 0$ )       | -0.013 | 0.130  | 0.314  | 0.570 |
| Transformer                    | -0.226 | -0.043 | 0.191  | 0.503 |
| Transformer, all masks         | -0.221 | -0.038 | 0.190  | 0.486 |
| Residual MLP                   | -0.107 | 0.039  | 0.235  | 0.508 |
| Scalar-cavity network          | 0.133  | 0.256  | 0.416  | 0.635 |
| Lifted cavity transformer (v3) | -0.196 | -0.037 | 0.179  | 0.482 |

Table 15: Panel F5 (linear sensors; 128 tasks): mean test NLL by number of missing sensors.

| Method                         | $k=0$  | $k=1$  | $k=2$  | $k=3$ |
|--------------------------------|--------|--------|--------|-------|
| Oracle                         | -0.455 | -0.288 | -0.066 | 0.244 |
| FA-Gaussian                    | -0.362 | -0.203 | 0.009  | 0.305 |
| EM-Gaussian                    | -0.244 | -0.142 | 0.034  | 0.308 |
| NL-FA                          | -0.337 | -0.178 | 0.030  | 0.318 |
| BLR                            | -0.135 | -0.085 | 0.061  | 0.322 |
| Ridge (complete case)          | -0.169 | -0.092 | 0.062  | 0.323 |
| Lifted cavity (ours)           | -0.323 | -0.172 | 0.031  | 0.318 |
| Lifted cavity, $K = 2$         | -0.320 | -0.170 | 0.033  | 0.318 |
| Lifted sites, no cavity        | -0.343 | -0.189 | 0.018  | 0.309 |
| Scalar sites ( $K = 0$ )       | -0.085 | 0.031  | 0.186  | 0.415 |
| Transformer                    | -0.278 | -0.122 | 0.085  | 0.371 |
| Transformer, all masks         | -0.281 | -0.123 | 0.080  | 0.353 |
| Residual MLP                   | -0.304 | -0.161 | 0.036  | 0.319 |
| Scalar-cavity network          | 0.052  | 0.161  | 0.302  | 0.505 |
| Lifted cavity transformer (v3) | -0.319 | -0.171 | 0.029  | 0.315 |

Table 16: Panel F6 (16 support rows; 128 tasks): mean test NLL by number of missing sensors.

| Method                         | $k=0$  | $k=1$  | $k=2$  | $k=3$ |
|--------------------------------|--------|--------|--------|-------|
| Oracle                         | -0.446 | -0.280 | -0.060 | 0.252 |
| FA-Gaussian                    | 0.256  | 0.371  | 0.510  | 0.686 |
| EM-Gaussian                    | 24.900 | 4.682  | 1.140  | 0.739 |
| NL-FA                          | 0.314  | 0.388  | 0.496  | 0.661 |
| BLR                            | 0.898  | 0.857  | 3.690  | 0.887 |
| Ridge (complete case)          | 0.317  | 0.391  | 0.500  | 0.624 |
| Lifted cavity (ours)           | 0.073  | 0.195  | 0.363  | 0.604 |
| Lifted cavity, $K = 2$         | 0.156  | 0.250  | 0.392  | 0.614 |
| Lifted sites, no cavity        | 0.117  | 0.234  | 0.389  | 0.604 |
| Scalar sites ( $K = 0$ )       | 0.204  | 0.293  | 0.428  | 0.644 |
| Transformer                    | 0.253  | 0.371  | 0.523  | 0.731 |
| Transformer, all masks         | 0.256  | 0.359  | 0.483  | 0.653 |
| Residual MLP                   | 0.094  | 0.211  | 0.371  | 0.594 |
| Scalar-cavity network          | 0.345  | 0.439  | 0.565  | 0.750 |
| Lifted cavity transformer (v3) | 0.237  | 0.333  | 0.457  | 0.643 |

Table 17: Panel F7 (96 support rows; 128 tasks): mean test NLL by number of missing sensors.

| Method                         | $k=0$  | $k=1$  | $k=2$  | $k=3$ |
|--------------------------------|--------|--------|--------|-------|
| Oracle                         | -0.429 | -0.264 | -0.043 | 0.269 |
| FA-Gaussian                    | -0.232 | -0.082 | 0.114  | 0.385 |
| EM-Gaussian                    | -0.237 | -0.086 | 0.112  | 0.385 |
| NL-FA                          | -0.276 | -0.127 | 0.073  | 0.353 |
| BLR                            | -0.178 | -0.056 | 0.125  | 0.390 |
| Ridge (complete case)          | -0.163 | -0.047 | 0.129  | 0.391 |
| Lifted cavity (ours)           | -0.324 | -0.163 | 0.052  | 0.353 |
| Lifted cavity, $K = 2$         | -0.320 | -0.159 | 0.056  | 0.357 |
| Lifted sites, no cavity        | -0.275 | -0.123 | 0.078  | 0.359 |
| Scalar sites ( $K = 0$ )       | -0.097 | 0.034  | 0.205  | 0.443 |
| Transformer                    | -0.295 | -0.133 | 0.081  | 0.379 |
| Transformer, all masks         | -0.297 | -0.134 | 0.080  | 0.367 |
| Residual MLP                   | -0.287 | -0.131 | 0.074  | 0.360 |
| Scalar-cavity network          | 0.047  | 0.161  | 0.309  | 0.516 |
| Lifted cavity transformer (v3) | -0.329 | -0.167 | 0.047  | 0.346 |

Table 18: Air Quality CO: mean test NLL for every number of simulated missing sensors (23 later weeks).

| Method                                  | $k=0$  | $k=1$  | $k=2$  | $k=3$ |
|-----------------------------------------|--------|--------|--------|-------|
| EM-Gaussian                             | 0.074  | 0.063  | 0.085  | 0.146 |
| FA-Gaussian                             | 0.052  | 0.086  | 0.116  | 0.154 |
| NL-FA                                   | 0.144  | 0.120  | 0.091  | 0.124 |
| Bayesian linear regression              | 1.037  | 0.433  | 0.216  | 0.202 |
| Ridge (complete case)                   | 0.824  | 0.337  | 0.208  | 0.210 |
| GP, complete case <sup>†</sup>          | 3.364  | 1.056  | 0.503  | 0.146 |
| $k$ -nearest neighbours <sup>†</sup>    | 0.571  | 0.391  | 0.353  | 0.274 |
| Lifted cavity, zero-shot                | 0.044  | 0.064  | 0.129  | 0.160 |
| Transformer, zero-shot                  | -0.017 | -0.006 | 0.081  | 0.154 |
| <b>Lifted cavity, fine-tuned (ours)</b> | -0.040 | -0.031 | 0.008  | 0.077 |
| no synthetic pretraining                | -0.001 | -0.014 | -0.006 | 0.077 |
| no cavity input                         | -0.086 | -0.065 | -0.028 | 0.038 |
| scalar sites ( $K = 0$ )                | -0.026 | -0.021 | -0.003 | 0.075 |
| Transformer, fine-tuned                 | -0.163 | -0.149 | -0.052 | 0.036 |
| Residual MLP, fine-tuned                | 0.025  | 0.061  | 0.110  | 0.160 |

Table 19: Air Quality NO<sub>2</sub>: mean test NLL for every number of simulated missing sensors (23 later weeks).

| Method                                  | $k=0$  | $k=1$  | $k=2$  | $k=3$  |
|-----------------------------------------|--------|--------|--------|--------|
| EM-Gaussian                             | −0.237 | −0.174 | −0.175 | −0.168 |
| FA-Gaussian                             | −0.196 | −0.173 | −0.155 | −0.162 |
| NL-FA                                   | −0.090 | −0.102 | −0.142 | −0.169 |
| Bayesian linear regression              | −0.055 | −0.214 | −0.178 | −0.160 |
| Ridge (complete case)                   | −0.200 | −0.230 | −0.176 | −0.145 |
| GP, complete case <sup>†</sup>          | 25.217 | 0.696  | −0.198 | −0.258 |
| $k$ -nearest neighbours <sup>†</sup>    | −0.010 | −0.020 | −0.063 | −0.128 |
| Lifted cavity, zero-shot                | −0.350 | −0.295 | −0.233 | −0.176 |
| Transformer, zero-shot                  | −0.360 | −0.310 | −0.231 | −0.136 |
| <b>Lifted cavity, fine-tuned (ours)</b> | −0.412 | −0.366 | −0.337 | −0.273 |
| no synthetic pretraining                | −0.419 | −0.365 | −0.316 | −0.248 |
| no cavity input                         | −0.423 | −0.375 | −0.332 | −0.275 |
| scalar sites ( $K = 0$ )                | −0.359 | −0.337 | −0.302 | −0.231 |
| Transformer, fine-tuned                 | −0.485 | −0.429 | −0.381 | −0.294 |
| Residual MLP, fine-tuned                | −0.246 | −0.198 | −0.147 | −0.106 |

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